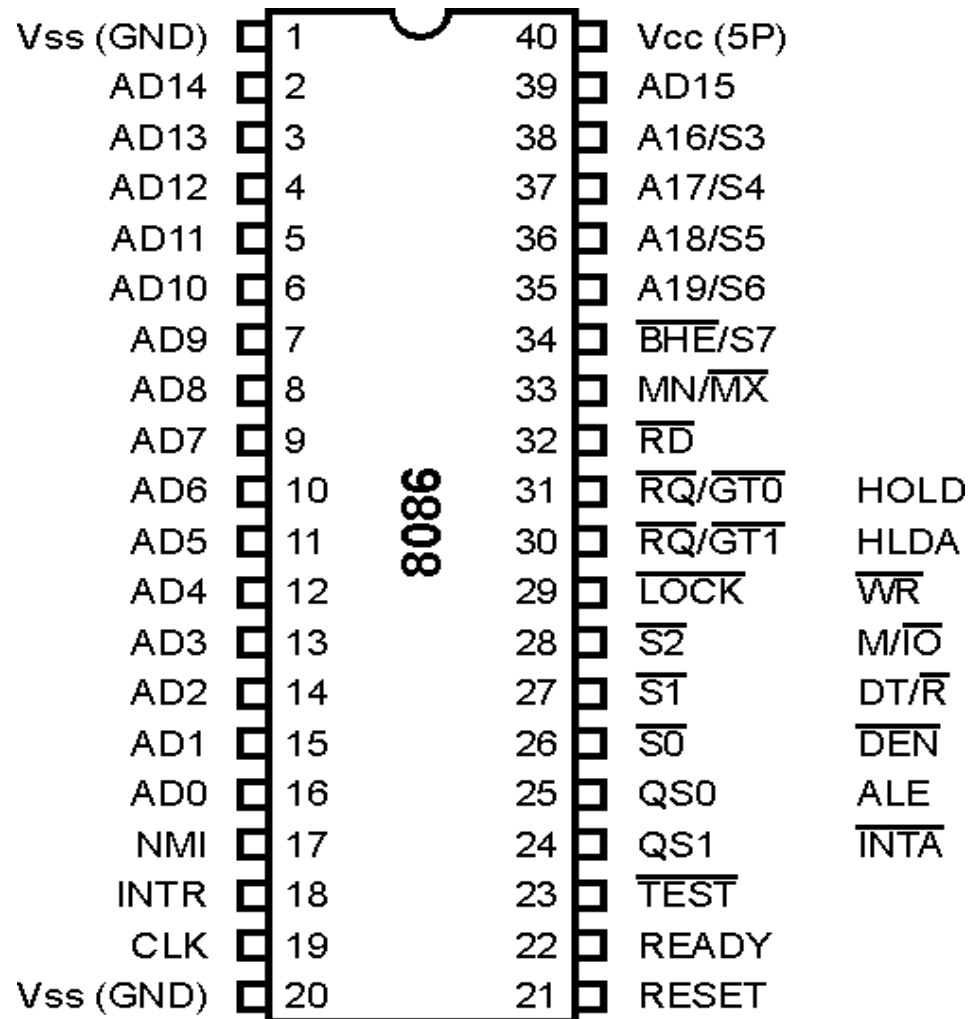


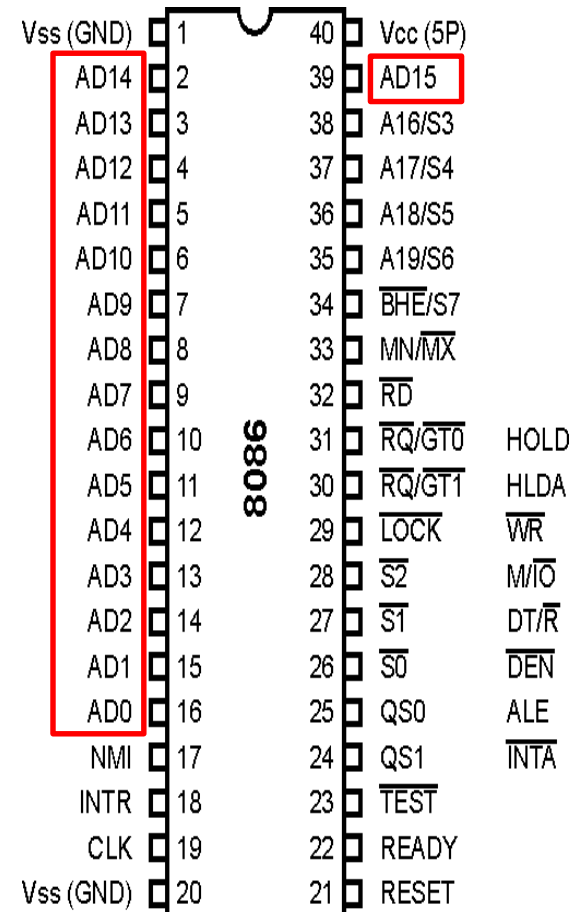
# Pin Diagram of Intel 8086



# AD<sub>0</sub> – AD<sub>15</sub>

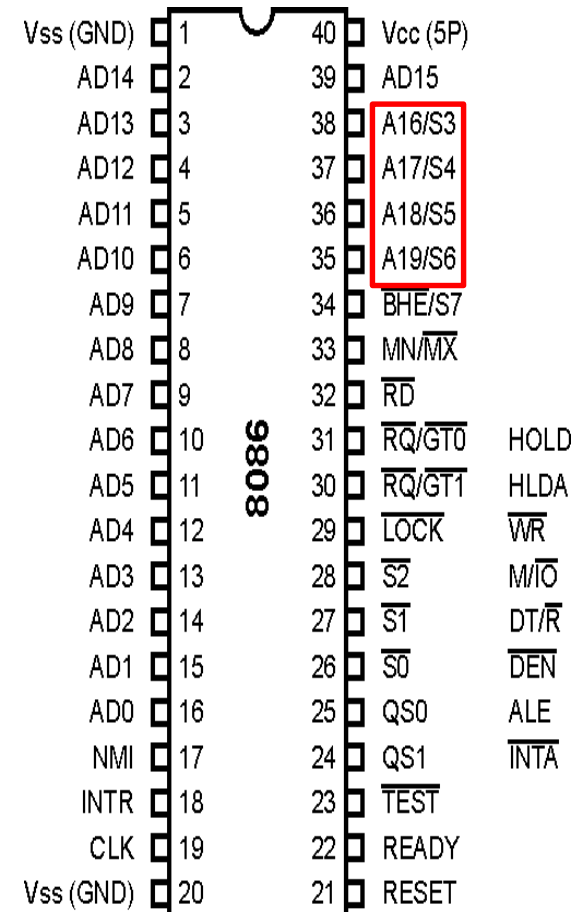
## Pin 16-2, 39 (Bi-directional)

- ❑ These lines are multiplexed bi-directional address/data bus.
- ❑ During T<sub>1</sub>, they carry lower order 16-bit address.
- ❑ In the remaining clock cycles, they carry 16-bit data.
- ❑ AD<sub>0</sub>-AD<sub>7</sub> carry lower order byte of data.
- ❑ AD<sub>8</sub>-AD<sub>15</sub> carry higher order byte of data.



# $A_{19}/S_6, A_{18}/S_5, A_{17}/S_4, A_{16}/S_3$ Pin 35-38 (Unidirectional)

- These lines are multiplexed unidirectional address and status bus.
- During  $T_1$ , they carry higher order 4-bit address.
- In the remaining clock cycles, they carry status signals.



**$A_{19}/S_6, A_{18}/S_5, A_{17}/S_4, A_{16}/S_3$**   
Pin 35-38 (Unidirectional)

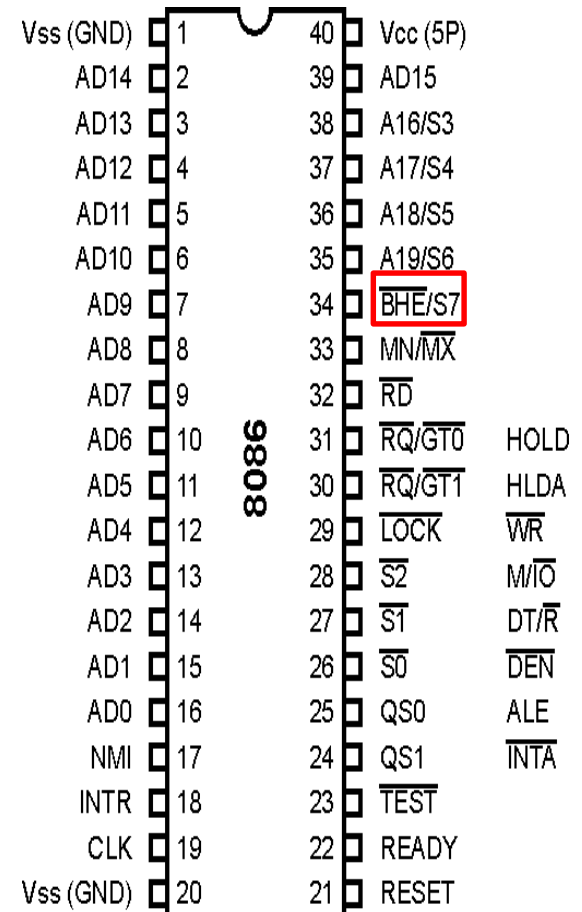
| A17/S4 | A16/S3 | FUNCTION             |
|--------|--------|----------------------|
| 0      | 0      | Extra segment access |
| 0      | 1      | Stack segment access |
| 1      | 0      | Code segment access  |
| 1      | 1      | Data segment access  |

- ❑  $S_5$  is interrupt enable
- ❑  $S_6$  is used as bus master, which handles the internal bus control. When Low, it indicates that 8086 is in control of the bus.

# BHE / S<sub>7</sub>

## Pin 34 (Output)

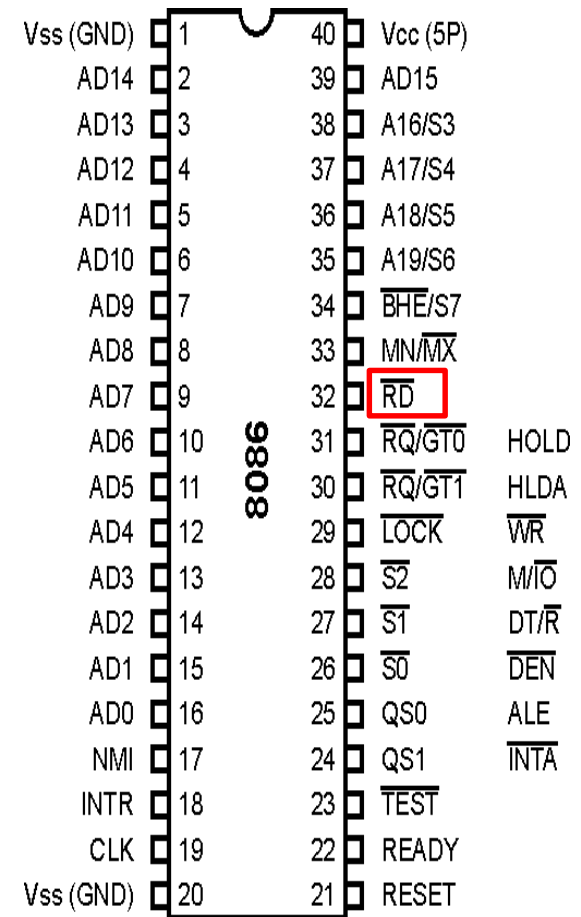
- ❑ BHE stands for Bus High Enable.
- ❑ BHE signal is used to indicate the transfer of data over higher order data bus ( $D_8 - D_{15}$ ).
- ❑ When BHE' is enabled (i.e. 0), then the bus will carry 16-bit data, else only 8-bit data through the lower order data bus lines.
- ❑ 8-bit I/O devices use this signal.
- ❑ It is multiplexed with status pin S<sub>7</sub>. S<sub>7</sub> signal is available during T<sub>2</sub>, T<sub>3</sub> and T<sub>4</sub>.



# RD (Read)

## Pin 32 (Output)

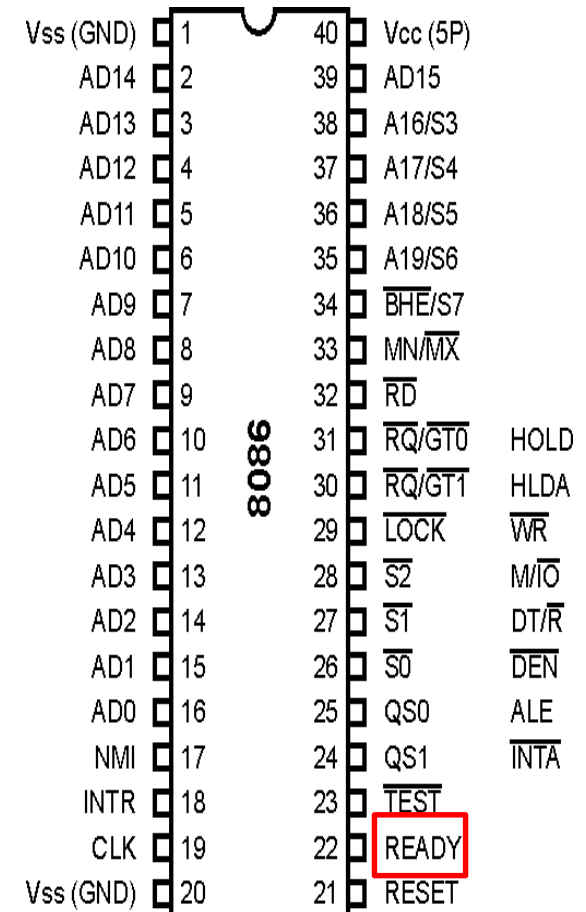
- ❑ It is a read signal used for read operation.
- ❑ It is an output signal.
- ❑ It is an active low signal.



# READY

## Pin 22 (Input)

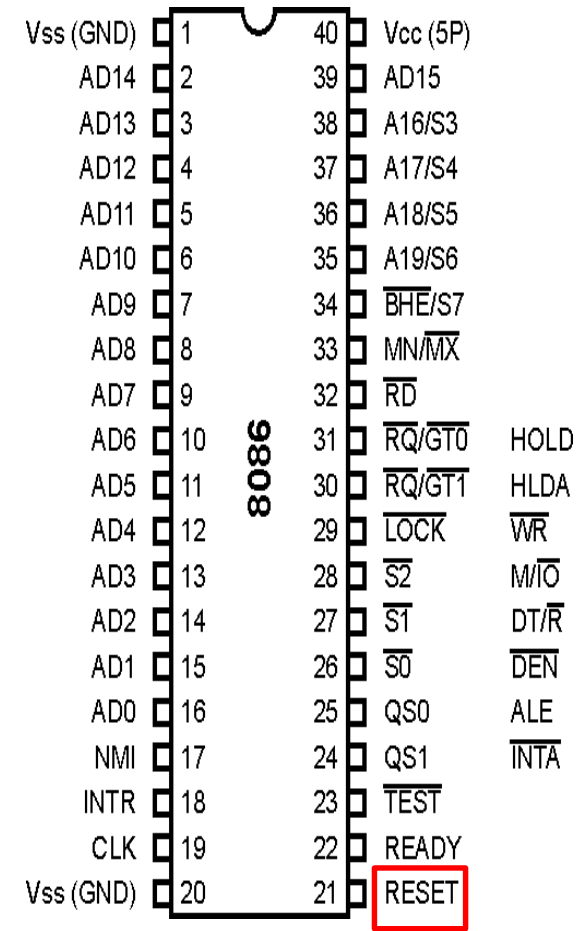
- ❑ This is an acknowledgement signal from slower I/O devices or memory.
- ❑ It is an active high signal.
- ❑ When high, it indicates that the device is ready to transfer data.
- ❑ When low, then microprocessor is in wait state.



# RESET

## Pin 21 (Input)

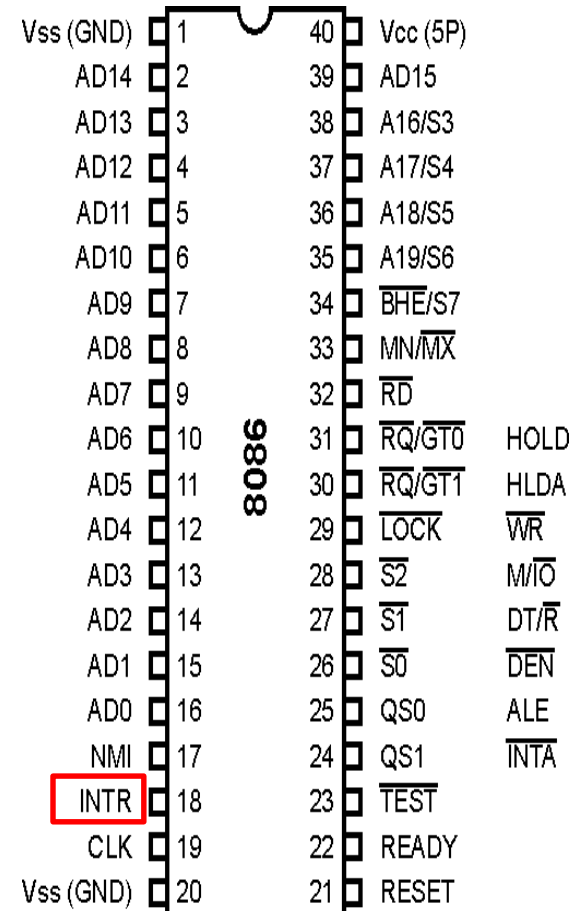
- It is a system reset.
- It is an active high signal.
- When high, microprocessor enters into reset state and terminates the current activity.
- It must be active for at least four clock cycles to reset the microprocessor.



# INTR

## Pin 18 (Input)

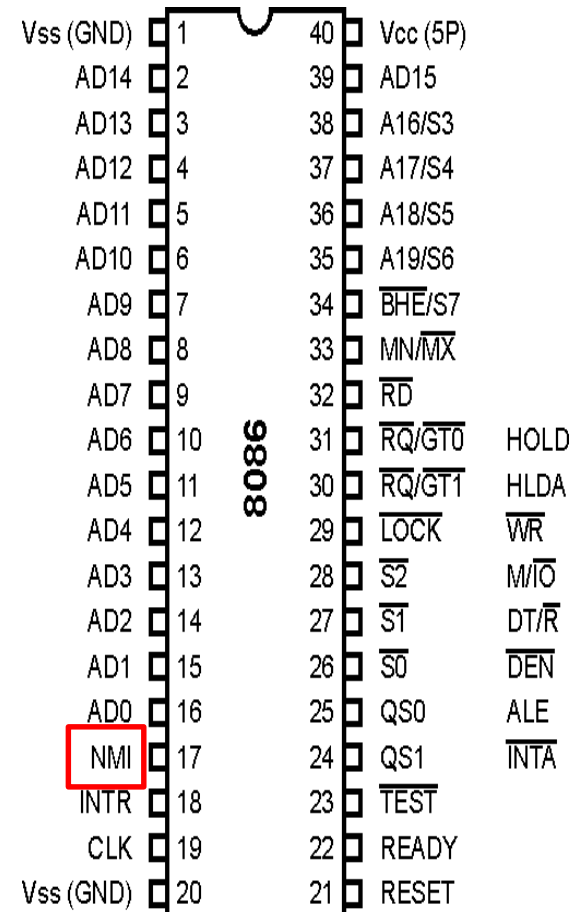
- It is an interrupt request signal.
- It is active high.
- It is level triggered.
- This is sampled during the last clock cycles of each instruction for determining the availability of the request.
- If any interrupt request is found pending, the processor enters the interrupt acknowledge cycle.



# NMI

## Pin 17 (Input)

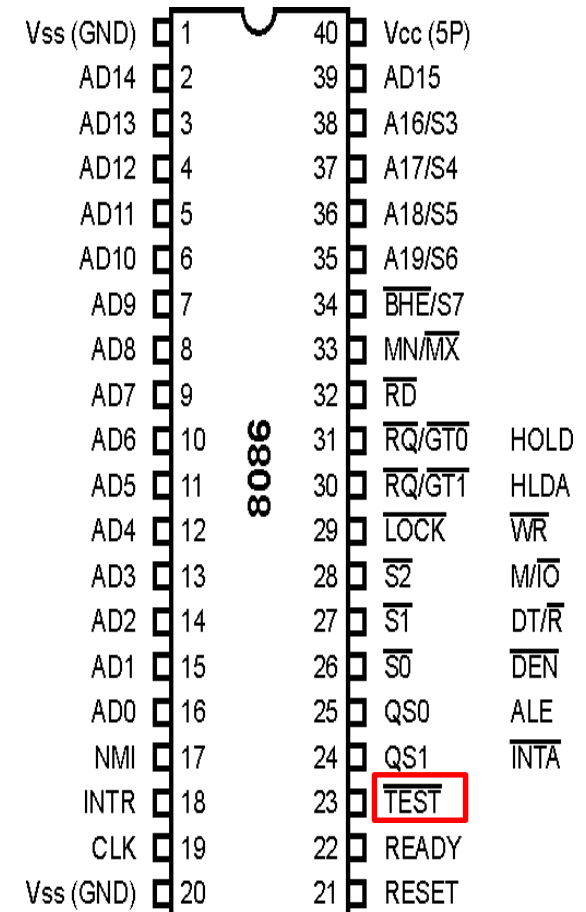
- It is a non-maskable interrupt signal.
- It is an active high.
- It is an edge triggered interrupt which results in a type II interrupt.
- Non Maskable **Interrupt**: The hardware which cannot be delayed and should process by the processor immediately



# TEST

## Pin 23 (Input)

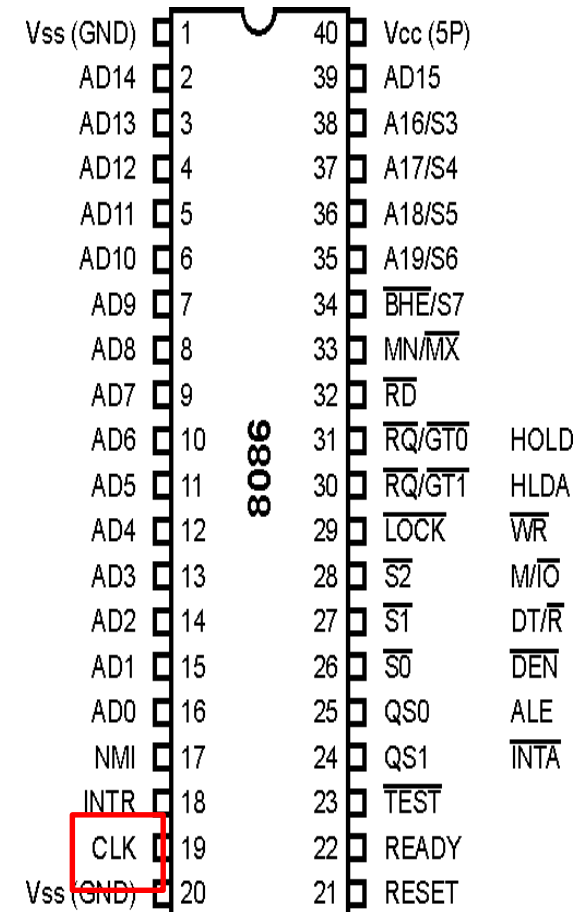
- It is used to test the status of math co-processor 8087.
- The BUSY pin of 8087 is connected to this pin of 8086.
- If low, execution continues else microprocessor is in wait state.



# CLK

## Pin 19 (Input)

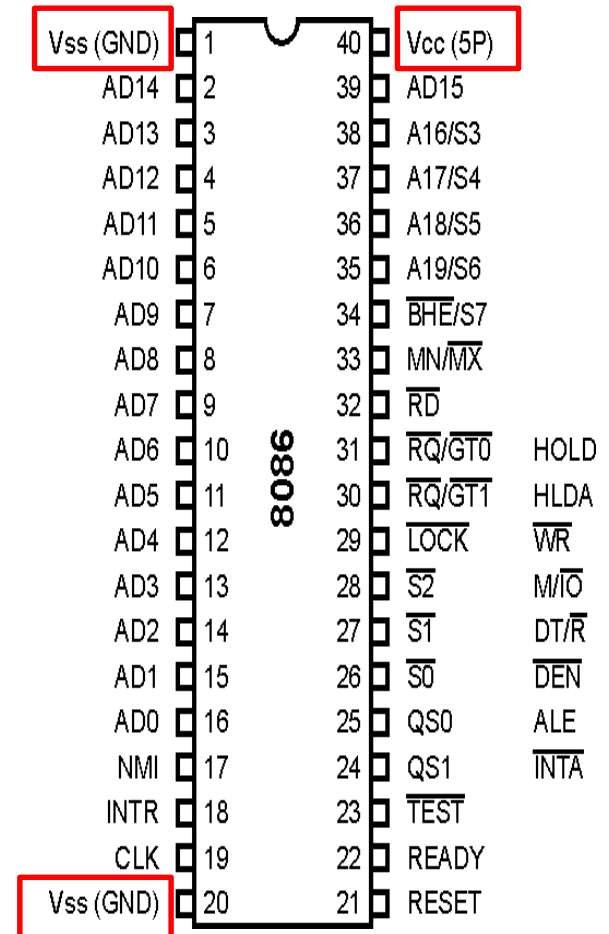
- This clock input provides the basic timing for processor operation.
- It is symmetric square wave with 33% duty cycle.
- The range of frequency of different versions is 5 MHz, 8 MHz and 10 MHz.



# $V_{CC}$ and $V_{SS}$

Pin 40 and Pin 20 (Input)

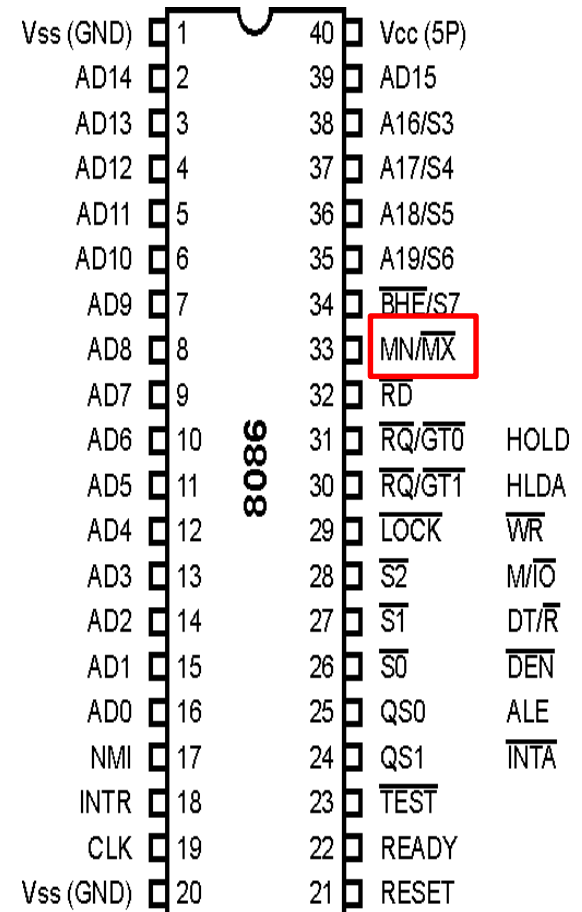
- $V_{CC}$  is power supply signal.
- +5V DC is supplied through this pin.
- $V_{SS}$  is ground signal.



# MN / $\overline{\text{MX}}$

## Pin 33 (Input)

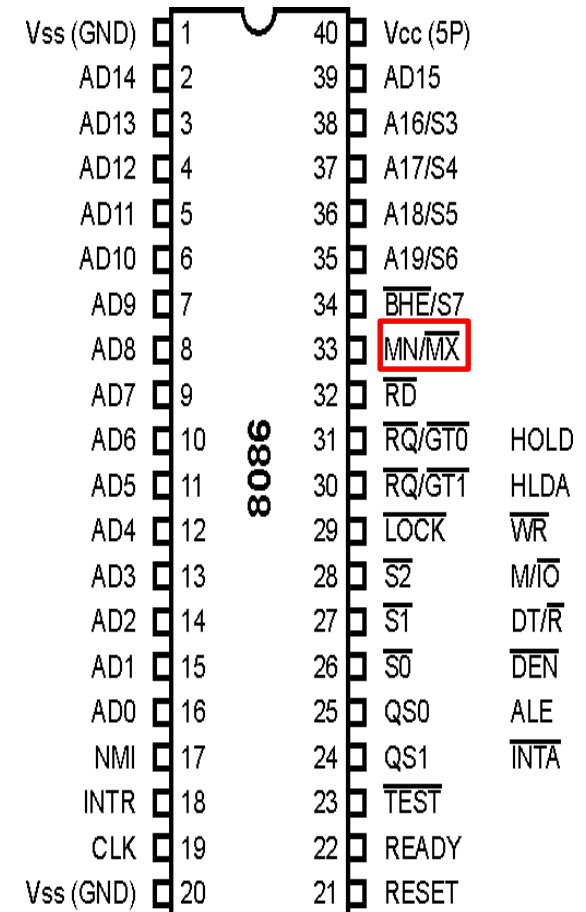
- 8086 works in two modes:
  - Minimum Mode
  - Maximum Mode
- If  $\text{MN}/\overline{\text{MX}}$  is high, it works in minimum mode.
- If  $\text{MN}/\overline{\text{MX}}$  is low, it works in maximum mode.
- In minimum mode there can be only one processor i.e. 8086.
- In maximum mode there can be multiple processors with 8086, like 8087 and 8089.



# MN / MX

## Pin 33 (Input)

- Pins 24 to 31 issue two different sets of signals.
- One set of signals is issued when CPU operates in minimum mode.
- Other set of signals is issued when CPU operates in maximum mode.

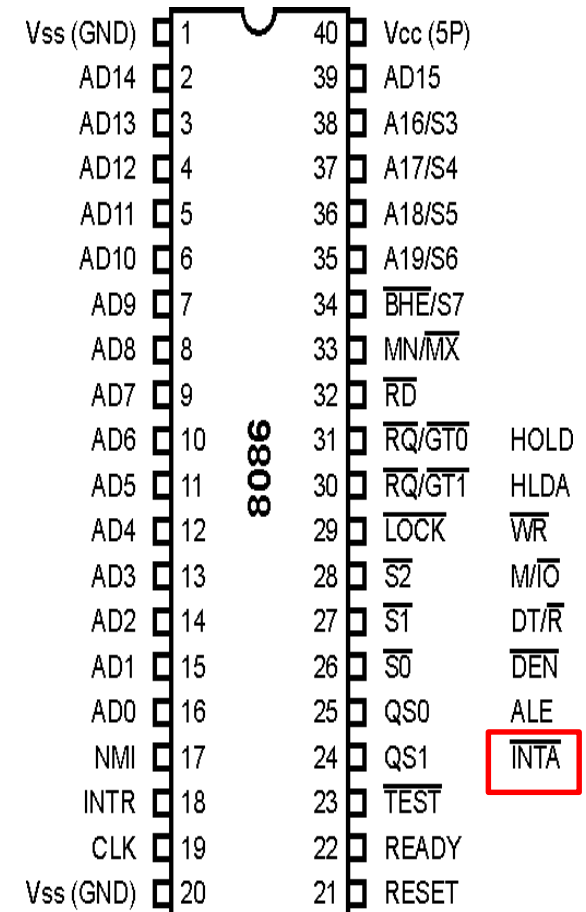


# Pin Description for Minimum Mode

# INTA

## Pin 24 (Output)

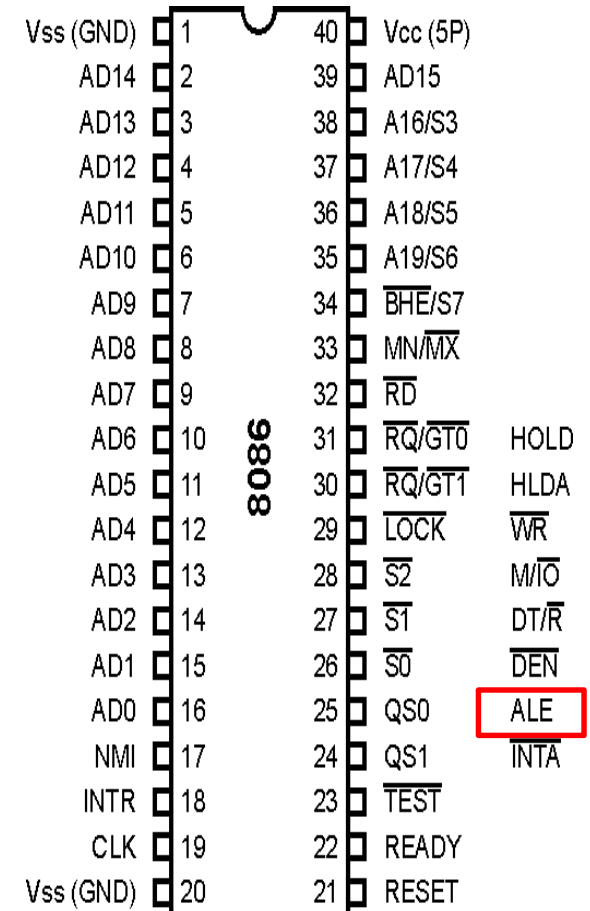
- This is an interrupt acknowledge signal.
- When microprocessor receives INTR signal, it acknowledges the interrupt by generating this signal.
- It is an active low signal.



# ALE

## Pin 25 (Output)

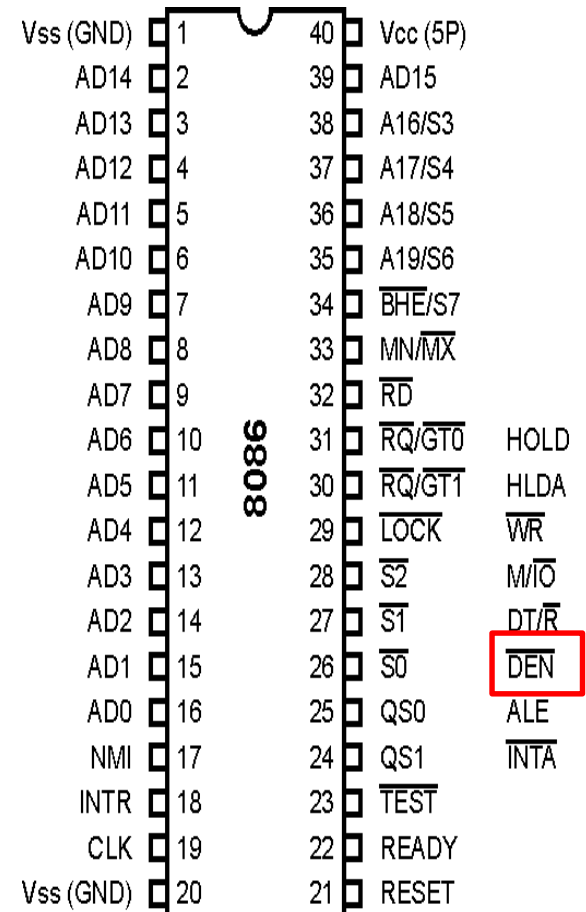
- This is an Address Latch Enable signal.
- It indicates that valid address is available on bus  $AD_0 - AD_{15}$ .
- It is an active high signal and remains high during  $T_1$  state.
- It is connected to enable pin of latch 8282.



# DEN

## Pin 26 (Output)

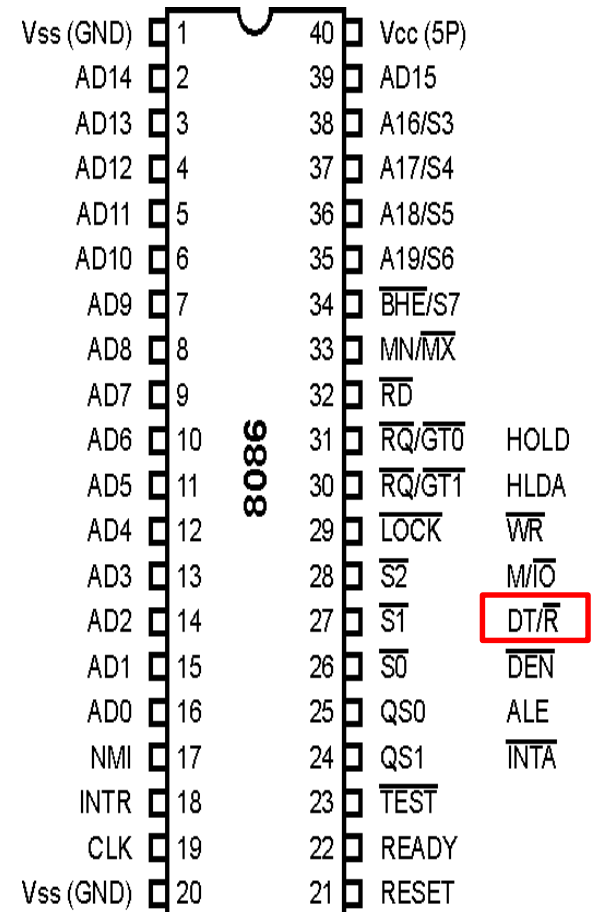
- This is a Data Enable signal.
- This signal is used to enable the transceiver 8286.
- Transceiver is used to separate the data from the address/data bus.
- It is an active low signal.



# DT / $\overline{R}$

## Pin 27 (Output)

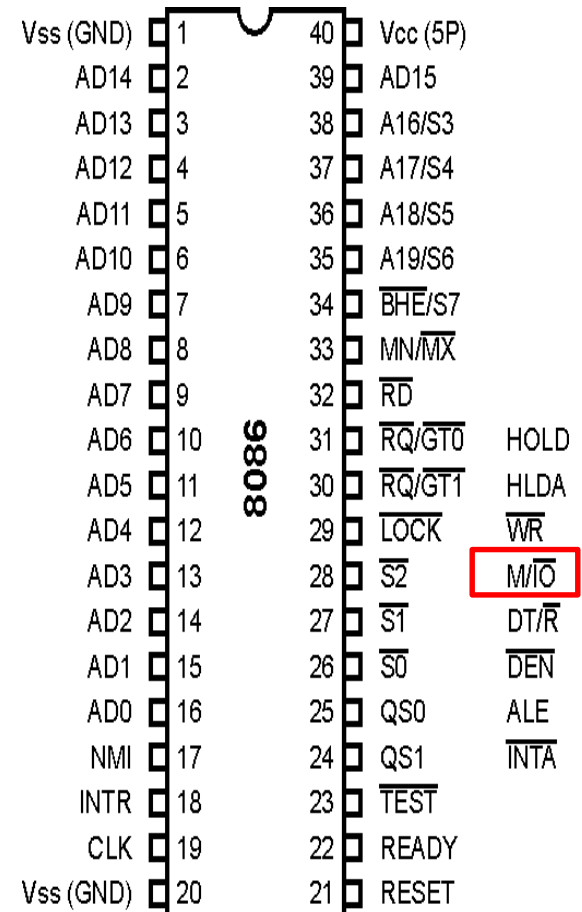
- This is a Data Transmit/Receive signal.
- It decides the direction of data flow through the transceiver.
- When it is high, data is transmitted out.
- When it is low, data is received in.



# M / $\overline{\text{IO}}$

## Pin 28 (Output)

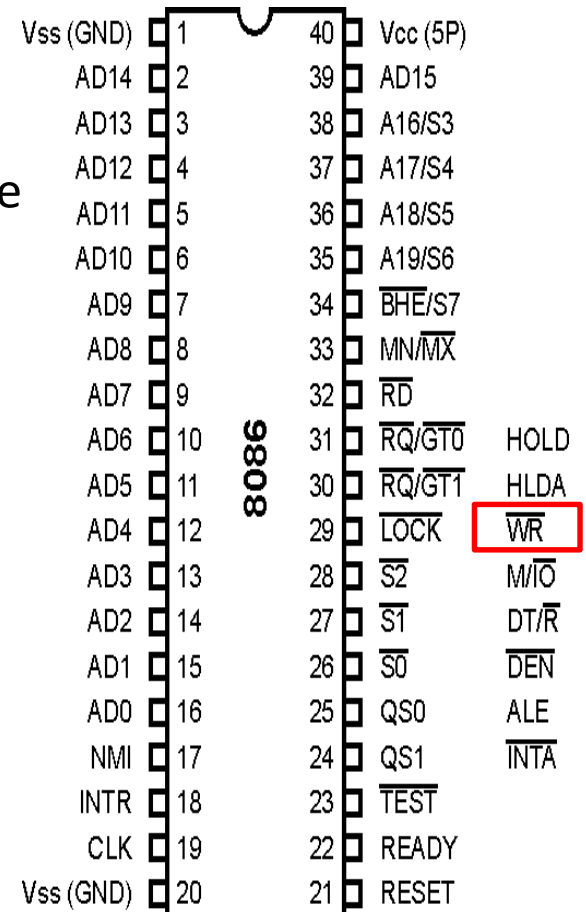
- This signal is issued by the microprocessor to distinguish memory access from I/O access.
- When it is high, memory is accessed.
- When it is low, I/O devices are accessed.



# WR

## Pin 29 (Output)

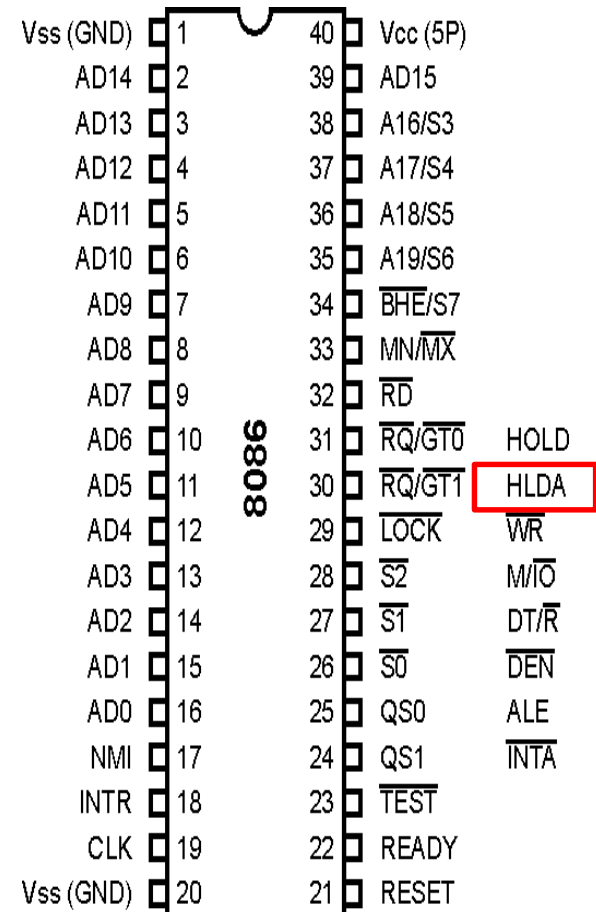
- It is a Write signal.
- It is used to write data in memory or output device depending on the status of  $\overline{M/\overline{IO}}$  signal.
- It is an active low signal.



# HLDA

## Pin 30 (Output)

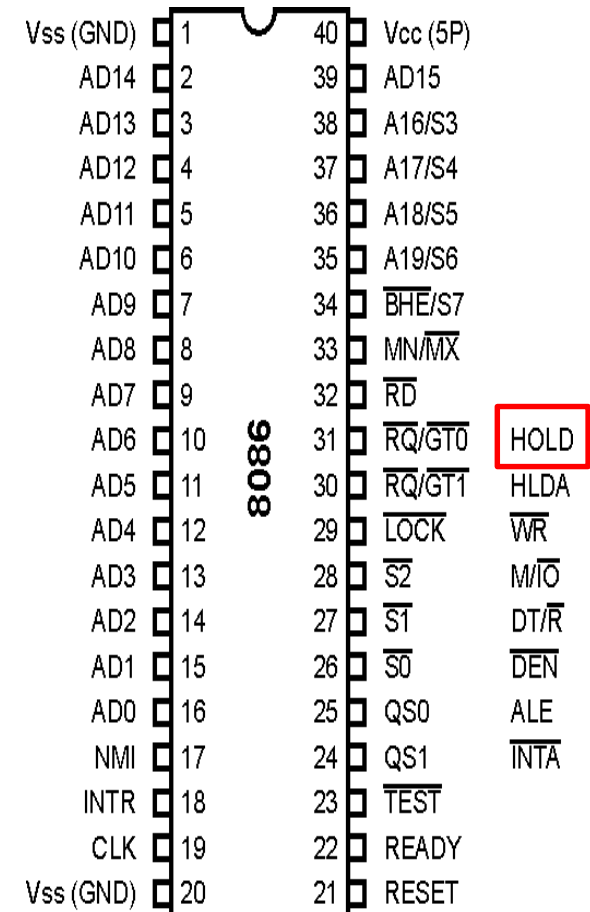
- It is a Hold Acknowledge signal.
- It is issued after receiving the HOLD signal.
- It is an active high signal.



# HOLD

## Pin 31 (Input)

- When DMA controller needs to use address/data bus, it sends a request to the CPU through this pin.
- It is an active high signal.
- When microprocessor receives HOLD signal, it issues HLDA signal to the DMA controller.

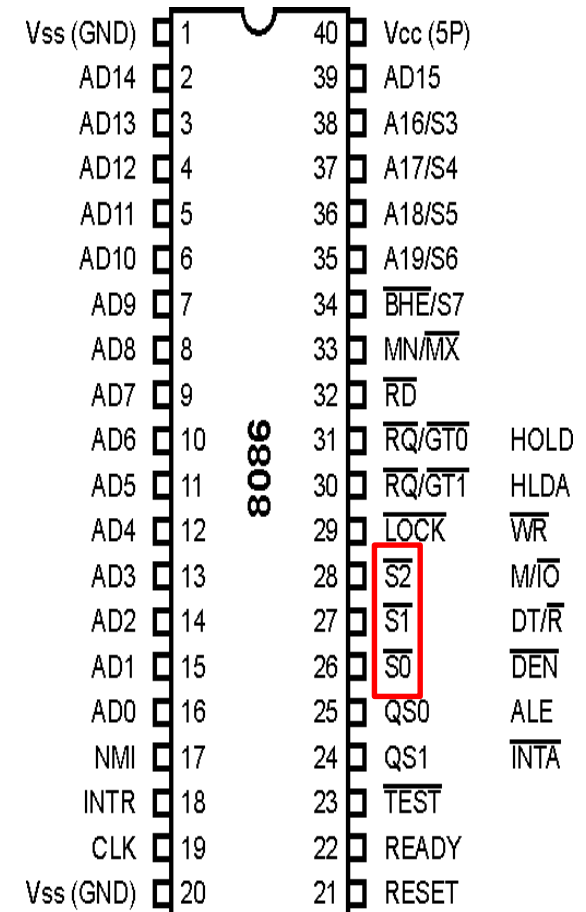


# Pin Description for Maximum Mode

# $\overline{S_0}, \overline{S_1}, \overline{S_2}$

Pin 26, 27, 28 (Output)

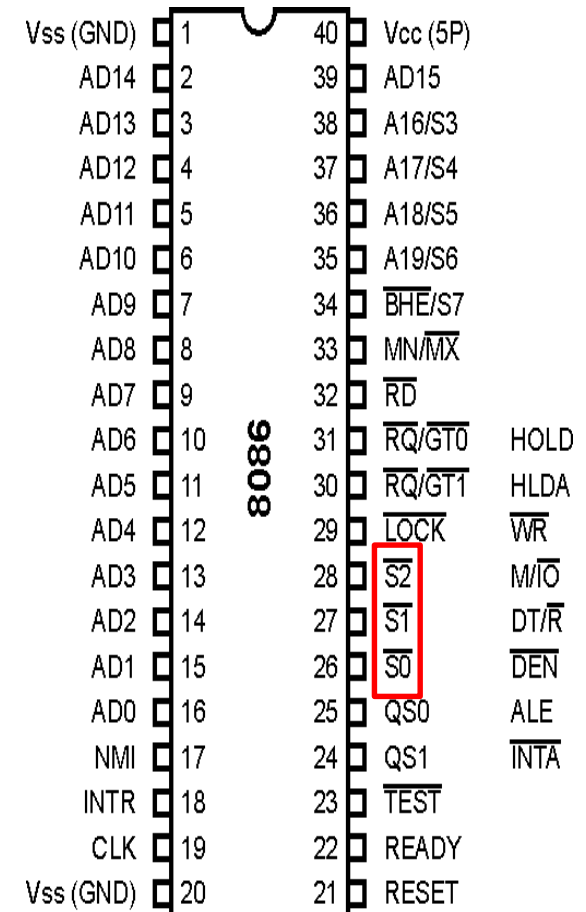
- These status signals indicate the operation being done by the microprocessor.
- This information is required by the Bus Controller 8288.
- Bus controller 8288 generates all memory and I/O control signals.
- These pins are active during T4, T1 and T2 states and is returned to passive state (1,1,1 during T3 or Tw (when ready is inactive).



# $\overline{S_0}, \overline{S_1}, \overline{S_2}$

Pin 26, 27, 28 (Output)

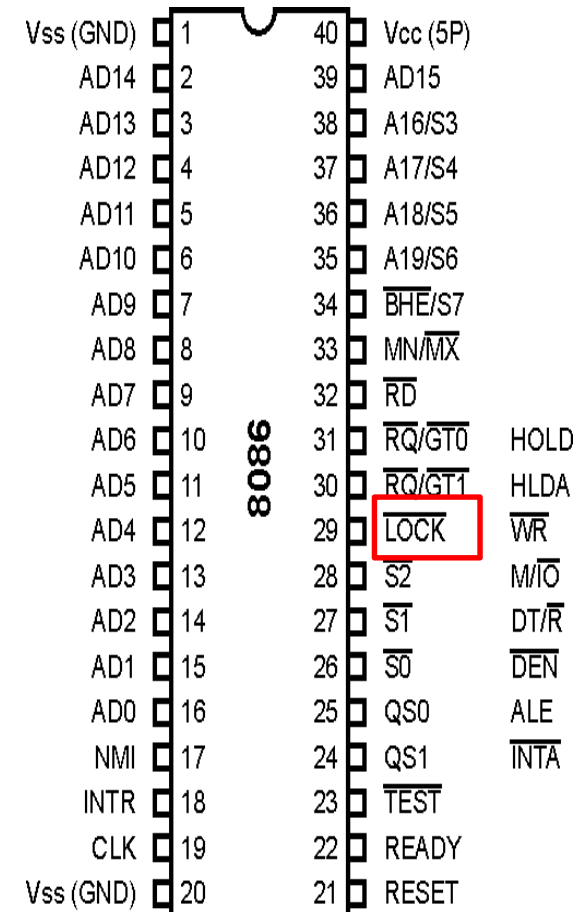
| $\overline{S_2}$ | $\overline{S_1}$ | $\overline{S_0}$ | Status                |
|------------------|------------------|------------------|-----------------------|
| 0                | 0                | 0                | Interrupt Acknowledge |
| 0                | 0                | 1                | I/O Read              |
| 0                | 1                | 0                | I/O Write             |
| 0                | 1                | 1                | Halt                  |
| 1                | 0                | 0                | Opcode Fetch          |
| 1                | 0                | 1                | Memory Read           |
| 1                | 1                | 0                | Memory Write          |
| 1                | 1                | 1                | Passive State         |



# LOCK

## Pin 29 (Output)

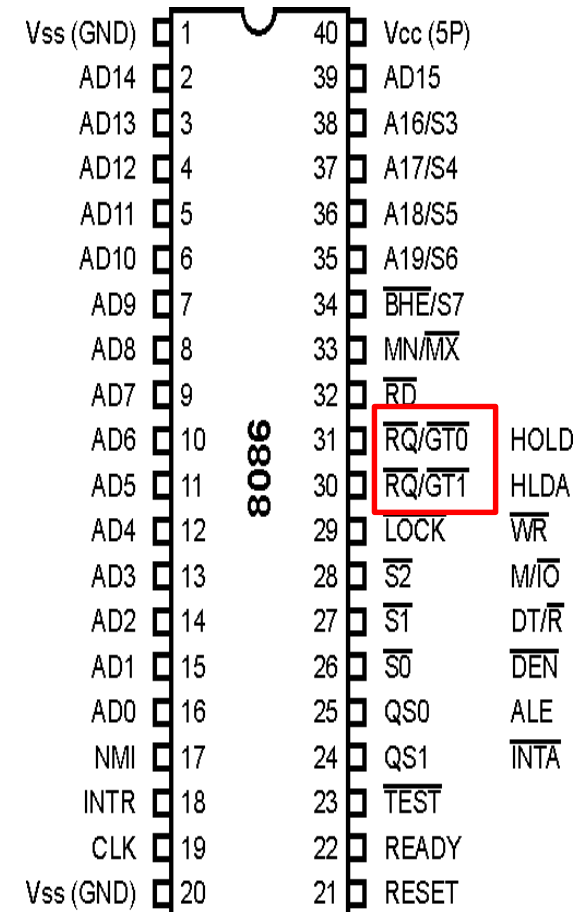
- This signal indicates that other processors should not ask CPU to relinquish the system bus.
- When it goes low, all interrupts are masked and HOLD request is not granted.
- This pin is activated by using LOCK prefix on any instruction.



# $\overline{RQ}/\overline{GT}_1$ and $\overline{RQ}/\overline{GT}_0$

Pin 30 and 31 (Bi-directional)

- These are Request/Grant pins.
- Other processors request the CPU through these lines to release the system bus.
- After receiving the request, CPU sends acknowledge signal on the same lines.
- $\overline{RQ}/\overline{GT}_0$  has higher priority than  $\overline{RQ}/\overline{GT}_1$ .

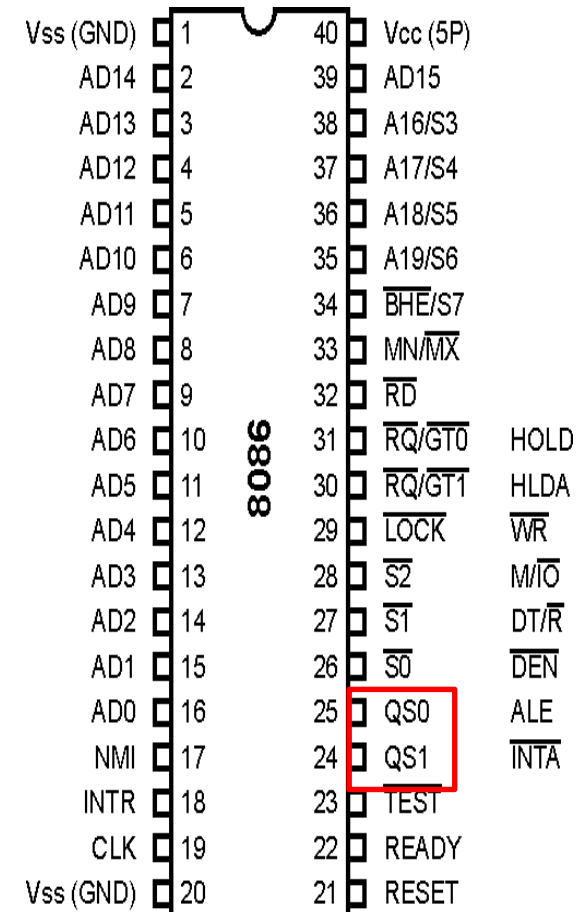


# QS0 and QS1

Pin 24 and Pin 25  
(Output)

- Queue Status. These signals indicate the status of the internal 8086 instruction queue according to the table shown below:

| QS1 | QS0 | STATUS                           |
|-----|-----|----------------------------------|
| 0   | 0   | No operation                     |
| 0   | 1   | First byte of op code from queue |
| 1   | 0   | Empty the queue                  |
| 1   | 1   | Subsequent byte from queue       |



**Thank You**